Plate bending with a circular hole - Validation in Kratos with IGA element

Objective:

Validate the Plate bending with a circular hole with Kratos I GA Shell3pElement for linear analysis.

Validation is performed against Michael Breitenberger's Dissertation – Examples – 7.3

1. Model Description:

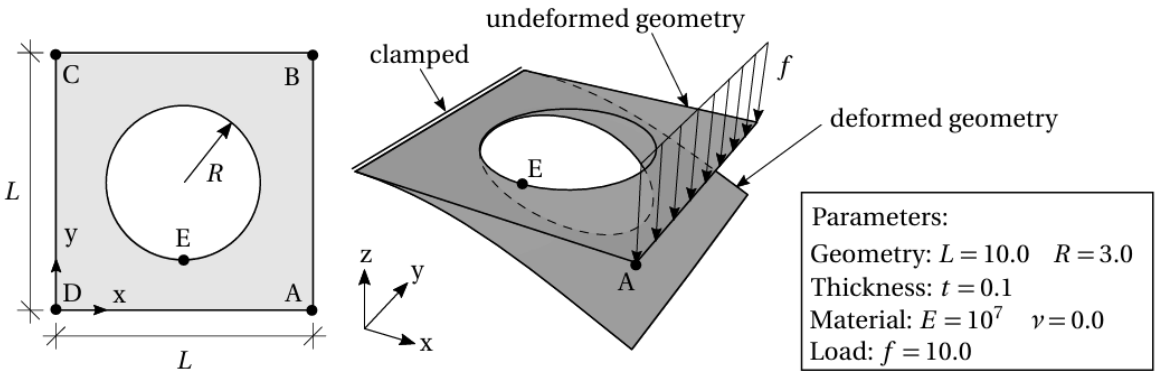

Fig.1: Plate with a circular hole in Bending - Taken from Michael Breitenberger's Dissertation

2. Results:

The Z displacement convergence value of -6.4313 units is achieved.

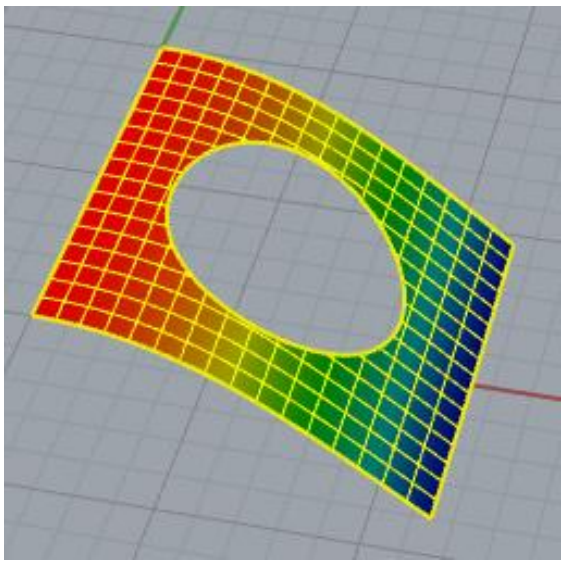

Fig.2: Deflection Diagram for CP of 15*15

In the convergence plot, the number of control points in the x and y directions is kept the same. The convergence plots shows number of control points added additionally with a curve degree of 2 in both directions.

Control Points Refinement	Z-Displacement
2	-6.2753
5	-6.4006
10	-6.431
15	-6.431

Table 1: Displacement Convergence Values

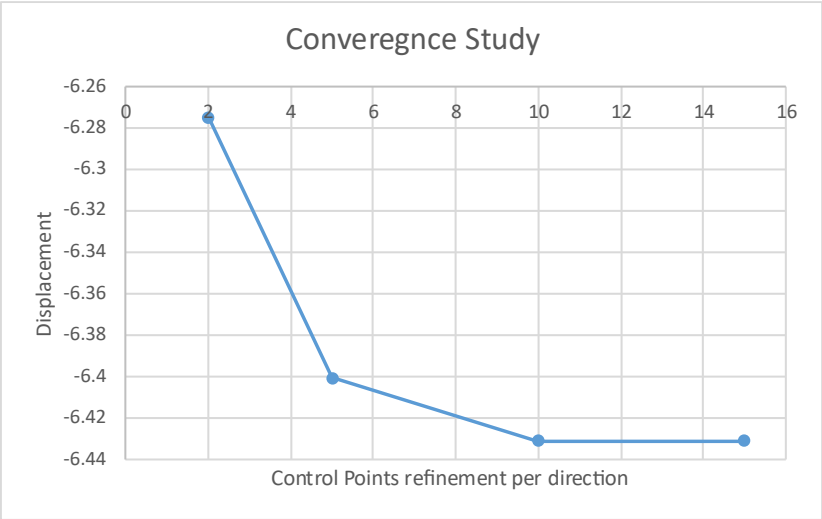

Figure 3: Convergence Plot